

Chapter 5

1. What is the purpose of a loop structure?

Loop structures are used to repeatedly prompt the user for a value until a certain condition is met. It can also be used for summing numbers or processing data.

2. Explain the difference between a while statement and a do-while statement.

With a do-statement the code executes at least once since it is not evaluated until after the loop runs. However with a while statement, it checks the condition before the loop runs. If the condition is false, then it may not even run.

3. An input validation loop is a loop that checks user input for valid data. If valid data is not entered, the loop iterates until valid data is entered. In which review of this chapter did you write code for an input validation loop.

The prompter review

4.

a) What is an infinite loop?

A loop that continues forever because its condition never becomes false.

b) List two types of errors that can lead to an infinite loop.

1. Syntax errors are common infinite loops.
2. When the loop condition never becomes false

c) What is meant by overflow?

It is when there are not enough bits to store a number. This can cause the output to be incorrect. An overflow makes the code function improperly.

5. How many times will the do-while loop execute?

```
int x = 0;
do {
    x = x + 2;
} while (x < 120);
```

The code can execute a maximum of 60 times. This is because the value of x can be anything 0 - 120. However since it is plus 2, that means only a maximum of 60 if it is every even number all the way to 120.

6. What initial value of x would make the loop infinite?

```
do {  
    x = x - 3;  
} while (x <120);
```

X can be any value under 120 to make the loop infinite. This is because the condition means if x is less than 120. Any value under 120 makes this condition true.

7. Compare and contrast counters and accumulators. List two uses for each.

An accumulator is a variable that increases by different amounts. A counter is a variable that increases by a fixed value. A counter for instance, counts how many times a number was entered. However with an accumulator, it can calculate the sums of numbers. It can also calculate the total score on a test.

8. Write a for statement that sums the integers from 3 to 10, inclusive.

```
int sum = 0;  
  
for (int i = 3; i <= 10; i++)  
{  
    sum = sum + i;  
}
```

9. List two factors that should be considered when determining which loop structure to choose

If a loop has to be run at least once, then a do-while loop is best fit. However, if you know the amount of times the loop needs to run, then use for-loop.

11. Consider the following assignment:

String x = "my string";

Determine the value returned by each of the following methods:

- a) `x.length()` 9
- b) `x.substring(0, 3)` "my "
- c) `x.toLowerCase()` "my string"
- d) `x.toUpperCase` "MY STRING"
- e) `x.trim()` "my string"